Options AI Trader

أدوات أفضل لنفس الإشارات + ما الذي يسبق حركة 5% في يومين؟

سبريد مدين · عقود طويلة · 64 مؤشرًا · 45 سهمًا 1999–2026 · أسعار عقود حقيقية

19 أكتوبر 2026 · الفرع rebuild/hybrid-engines-20261018

Alpaca 5 دقائق 2016–2026 · أسعار عقود حقيقية 2024–2026 · 15 رمزًا

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الملخص التنفيذي

يغطي هذا التقرير عملين طلبتهما:

الأول (الجولة 9): هل تنقذ أداة أفضل الإشارات الجيدة؟

- اختبرنا 5 إشارات أعطت حركة سهم إيجابية في الجولات السابقة، بأربع أدوات لكل صفقة:
  - **السهم نفسه.**
  - **Call أسبوعي.**
  - **سبريد مدين أسبوعي.**
  - **عقد طويل** (45–90 يومًا، دلتا 0.75).
- البيانات: أسعار عقود حقيقية 02-2024 → 10-2026، على الأسهم الأصلية والجديدة.
- معايير القبول سُجّلت مسبقًا قبل التشغيل.

الثاني (الجولة 10، فكرتك): ما الذي يسبق صعود السهم 5% أو أكثر خلال يومين؟

- درسنا 64 مؤشرًا ومجموعاتها على 45 سهمًا منذ 1999.
- بنينا استراتيجية من أفضل تركيبة، ثم اختبرناها:
  - على الأسهم كأسهم: 1999–2026.
  - بالخيارات بنموذج معايَر: 1999–2026.
  - بأسعار عقود حقيقية: 2024–2026.

أهم النتائج

النتيجة	التفاصيل
✅ **فكرتك أنتجت أفضل استراتيجية حتى الآن: "ارتداد ما بعد الانهيار"**	سهم هبط > 10% في 20 يومًا، وتقلبه مرتفع (عرض بولنجر في أعلى 20%)، والسوق هبط > 3% في 5 أيام ← شراء Call عند 15:50 والخروج بعد يومين
↳ على السهم 1999–2026	**+1.14% لكل صفقة**، 2,935 صفقة، 59% رابحة. موجبة في 26 من 27 سنة. **t = 6.8** على 129 موجة هبوط مستقلة
↳ Call 26–2024 أسبوعي، أسعار حقيقية	**+19.5%** (الأسهم الأصلية، 64 صفقة) و**+18.6%** (الأسهم الجديدة، 106 صفقات). مع فروق مضاعفة: +13% / +11.7%
⚠️ لكن	الصفقات الحقيقية تتركز في 3 موجات هبوط فقط (أغسطس 2024، ومارس–أبريل 2025). بدون أفضل 10 صفقات: +4.1%
🟡 **أسبوعي Call اليومي بـ Squeeze**	الوحيد الذي اجتاز معايير القبول المسجّلة مسبقًا في الجولة 9: +14.3% (107 صفقات، t = 1.51). لكن بدون أفضل 5 صفقات: +1.0%
❌ **السبريد المدين والعقد الطويل لم يُنقذا شيئًا**	السبريد أسوأ في كل الحالات، لأنه يدفع فرق سعرين. والعقد الطويل ≈ صفر
💡 **قوي جدًا على السهم (زخم 12 شهرًا) S1 نفسه**	+2.6% (الأصلية) و**+3.4%** (الجديدة) لكل شهر احتفاظ، t = 4–5. لكن العقد الطويل لم يلتقط هذا الربح (≈ 0%)

الاستنتاج العملي:

- **(1)** الحافة الحقيقية الوحيدة القابلة للتداول بالخيارات حتى الآن هي **شراء Call أسبوعي بعد انهيار حاد في سهم مع هبوط السوق**. هي حافة "انعكاس قصير المدى"، موثقة في الأدبيات ومتسقة عبر 27 سنة.
- **(2)** إشارات الزخم الطويل (S1) مربحة **على السهم** لا على الخيارات. إن سمحت بإضافة تداول الأسهم للتطبيق، فهي الأقوى إحصائيًا.
- **(3)** فرضيتي بأن "السبريد والعقود الطويلة أفضل" **لم تتأكد** بأسعار حقيقية. تكلفة تنفيذ ساقين أكبر من فائدة تقليل علاوة التقلب.

الجولة 10: ما الذي يسبق حركة ≥ 5% خلال يومين؟

المنهجية: لماذا "الرفع" لا "التكرار"

لو بحثنا فقط عن المؤشرات الموجودة قبل الحركات الكبيرة، فسنجد مؤشرات شائعة في كل الأيام. لذلك قسنا:

الرفع = P(حركة كبيرة | المؤشر) ÷ P(حركة كبيرة في أي يوم)

العنصر	التفاصيل
الحدث	إغلاق اليوم t+2 ÷ إغلاق اليوم (Call) +5% ≥ t − 1 أو ≥ −5% (Put). نسخة ثانية بالتقلب: ≤ ATR 2
المؤشرات (64)، كلها عند إغلاق اليوم t	؛ انضغاطADX؛ Stochasticو RSI2و RSI14 ؛ المتوسطات (20/50/200، الترتيب، الميل، العبور)؛(موجب، صاعد، تقاطع) MACD ؛ موقع الإغلاق؛ القرب من قمة/قاع 52 أسبوعًا؛ العوائد السابقة؛ القوةOBVوالشمعة الداخلية؛ الحجم و NR7 ؛ATR% واتساع بولنجر؛ ؛ **رقم موجة إليوت اليومية**؛ يوم الأسبوعVIXو SPY النسبية؛ اتجاه
الاكتشاف	الأسهم الـ 15 الأصلية، 1999–2015. البحث عن التركيبات تم هنا فقط، واشترطنا ثباتها في نصفي الفترة (1999–2007 و2008–2015)
الاختبار	الأسهم الأصلية 2016–2026، و**30 سهمًا جديدًا** 1999–2026

النتائج: المؤشرات المنفردة (صعود ≤ 5%)

النسبة الأساسية: 6.0% من الأيام في الاكتشاف، و4.6% في التحقق والأسهم الجديدة.

المؤشر	الرفع: اكتشاف	تحقق	أسهم جديدة	متوسط عائد يومين
عائد 20 يومًا < −10%	**2.94**	3.05	3.05	+0.36% / +0.48% / +0.41%
RSI14 < 30	**2.60**	2.74	2.41	+0.81% / +0.57% / +0.60%
عائد 5 أيام < −5%	**2.53**	2.74	2.70	+0.38% / +0.50% / +0.50%
هبط > 3% في 5 أيام SPY	**2.13**	2.31	2.52	+0.45% / +0.45% / +0.52%
VIX > 25	**2.06**	2.24	2.35	+0.30% / +0.36% / +0.31%
حجم > 2× المتوسط	**1.86**	1.98	1.72	+0.23% / +0.33% / +0.13%
إغلاق تحت بولنجر السفلي	**1.65**	1.60	1.59	+0.54% / +0.46% / +0.28%

المؤشرات "الإيجابية" التقليدية تُضعف احتمال الحركة الكبيرة.

المؤشر	الرفع
MACD > 0	0.75
SMA50 > SMA200	0.70
ترتيب EMA9 > EMA20 > EMA50	0.70

السبب: الأسهم في ترند هادئ تتحرك ببطء. **أرقام موجات إليوت لم تظهر في أي تركيبة ذات رفع ثابت للصعود.**

تنبيه مهم: نفس المؤشرات ترفع احتمال **الهبوط** الكبير أيضًا. مثلًا، عائد 20 يومًا < −10% يرفع احتمال الهبوط ≤ 5% بمعامل 2.73. أي أنها تتنبأ **بحجم** الحركة أكثر من اتجاهها.

لكن الاتجاه يميل للصعود: في أفضل تركيبة احتمال الصعود ≤ 5% ضعف احتمال الهبوط ≤ 5%.

المجموعة	صعود ≤ 5%	هبوط ≤ 5%
الاكتشاف	29%	15%
التحقق	22%	11%
الأسهم الجديدة	24%	13%

ومتوسط العائد +1.0% إلى +1.5%، و60% من الحالات موجبة.

أفضل تركيبة (المرتبة الأولى في الاكتشاف، ثم اختُبرت كما هي)

عرض بولنجر في أعلى 20% + عائد 20 يومًا < −10% + SPY هبط > 3% في 5 أيام

المجموعة	الأيام	الرفع	P(صعود ≤ 5) (%)	متوسط عائد يومين
اكتشاف 1999–2015	781	**4.84**	29%	+1.53%
تحقق 2016–2026	536	**4.77**	22%	+1.02%
أسهم جديدة 1999–2026	2,418	**5.28**	24%	+1.38%

نسخة ATR 2 (تُطبّع التقلب) أضعف خارج العينة. أفضل مؤشراتها: الحجم الكبير مع هبوط السوق.

كاستراتيجية: "ارتداد ما بعد الانهيار"

على 27 سنة (يومي، 45 سهمًا، الدخول عند الإغلاق والخروج بعد يومين):

المقياس	النتيجة
السهم	**+1.14% لكل صفقة**، 2,935 صفقة، 59% رابحة
سنوات موجبة	**26 من 27** (الاستثناء 2023: 9 صفقات فقط)
موجات الهبوط المستقلة	129؛ 73% منها موجبة؛ **t = 6.8** على مستوى الموجات
أسبوعي (نموذج معايَر) Call	+13.6% لكل صفقة؛ t = 3.4 على مستوى الموجات
أزمات صعبة	2008: سهم +1.7%، Call +22%؛ 2020: +0.3% / +6.6%؛ 2022: +1.0% / +3.9%؛ 2001: +0.2% / −1.7%

بأسعار عقود حقيقية 02-2024 → 10-2026:

الأداة	الأسهم الأصلية	الأسهم الجديدة
السهم	+2.4% (73، t 3.3)	+2.6% (155، t 6.1)
أسبوعي Call	**+19.5%** (64، t 1.71)	**+18.6%** (106، t 1.83)
فروق Call — 2×	+13.0%	+11.7%
سبريد مدين	+5.7%	+4.3%
سبريد — فروق 2×	−7.3%	−9.5%
عقد طويل	+4.6% (4 فقط)	−5.7% (11)

المتانة (Call، كل الأسهم، 170 صفقة):

الاختبار	النتيجة
الكل	+19.0%
بدون أفضل 5	+9.3%
بدون أفضل 10	+4.1%
أغسطس 2024	+12.0% (28)
مارس 2025	+14.1% (52)
أبريل 2025	+26.2% (77)
فبراير 2025	−55% (3)
نوفمبر 2025	−70% (5)
الأسهم / ETF	+20.2% / −0.7%

التقييم 🟡+ (أقوى مرشح حتى الآن): - **ما يدعمها:** الحافة في السهم ثابتة عبر 27 سنة و45 سهمًا. الخيارات بأسعار حقيقية موجبة على مجموعتين مستقلتين من الأسهم. - **ما يحدّ منها:** - الإشارات نادرة ومتجمعة: تظهر فقط في موجات

هبوط السوق، 2–4 مرات في السنة. - الربح يعتمد على صفقات كبيرة قليلة. - فترة الأسعار الحقيقية تحتوي 3 موجات فقط. - في هبوط طويل ممتد (مثل 2001) العائد قريب من الصفر.

القاعدة المقترحة (للتجربة في Paper)

```
CAPITULATION REBOUND CALL  (status: Paper candidate)
UNIVERSE  liquid US stocks with weekly options (ETFs showed no edge)
AT 15:50  (daily series whose close = 15:50 price)
  bbw_pct  = percentile rank of BB-width(20,2) over the last 126 days      > 0.80
  ret20    = close / close[20 days ago] - 1                                < -0.10
  spy_ret5 = SPY close / SPY close[5 days ago] - 1                         < -0.03
  -> BUY CALL, delta ~0.50, first Friday >= entry + 4 days, LIMIT at mid
EXIT      at 15:50 two sessions later (time exit only; no stop - the edge is a 2-day rebound)
SIZE      0.5-1 % of NAV premium per trade; max 5 positions; one per symbol;
          expect clusters: cap total premium per selloff episode at 5 % of NAV
SKIP      earnings within the holding window (when a calendar is available)
```

الجولة 9: أداة أفضل لنفس الإشارات؟

القواعد ومعايير القبول سُجّلت في `results/r9/PREREG.json` قبل التشغيل. القبول يتطلب كل ما يلي:

- عائد موجب على الأسهم الأصلية **و**الجديدة.
- t مجمّع > 1.
- عائد موجب في نصفي الفترة.
- عائد موجب مع فروق مضاعفة.

الإشارة	السهم (أصلية / جديدة)	أسبوعي Call	سبريد مدين	عقد طويل	الحكم
انضغاط يومي **SQZ**	+0.9% / +0.3%	**+19.4% / +11.9%**	−4.5% / −5.8%	+0.1% / −7.0%	**Call: مرشح Paper** (14.3+ مجمّع%، t 1.511.0+ ؛5: بدون أفضل %)
ارتداد الترند **TDL** (قادة فقط)	+0.4% / +0.3%	−4.2% / −10.5%	−9.9% / −15.0%	−2.6% / −10.5%	❌
FOMC	+0.2%	+4.2% (النصف الثاني −6.5%)	−3.2%	−0.5%	❌
زخم 12 شهرًا **S1**	**+2.6% / +3.4%** (t 4.0 / 5.0)	—	—	−1.3% / +0.3%	❌ بالخيارات، ✅ بالسهم
قمة 52 أسبوعًا **S3**	+0.8% / +0.4%	—	—	+4.6% / −3.8%	❌

لماذا لم يعمل السبريد والعقد الطويل؟

- **(1) السبريد:** يدفع نصف الفرق مرتين، لأنه ساقان. والساق المباعة تقصّ الربح في الحركات الكبيرة، وهي بالضبط ما يصنع ربح هذه الإشارات.
- **(2) العقد الطويل بدلتا 0.75:** أقل سيولة. طابقنا 40% فقط من صفقات S1 بتداول فعلي. وعلاوته الزمنية ما زالت تأكل حركة شهر عادي.
- **(3) S1:** أفضل إشارة لدينا على السهم (+2.6% إلى +3.4% شهريًا على مجموعتين مستقلتين). **أداتها الصحيحة هي السهم نفسه أو عقد شبه-سهم بدلتا 0.9.**

التوصيات

- **(1) أضف للتطبيق (Paper فقط):** "ارتداد ما بعد الانهيار" بـ Call أسبوعي، و"Squeeze اليومي" بـ Call أسبوعي، بحجم صغير. القواعد أعلاه.
- **(2) فكّر في السماح بتداول الأسهم:** لأن S1 وارتداد ما بعد الانهيار أقوى على السهم نفسه منهما على الخيارات. أو جرّب عقودًا بدلتا 0.9 كبديل للسهم.
- **(3) لا تُضف** السبريد المدين ولا العقود الطويلة لهذه الإشارات، فقد فشلت بأسعار حقيقية.
- **(4) لا تعتمد** أي شيء على أموال حقيقية قبل 60 صفقة Paper بأسعار التنفيذ الفعلية. ولاحظ أن "ارتداد ما بعد الانهيار" قد يحتاج سنة كاملة لجمع 60 صفقة بسبب ندرته.

> **تحفّظ:** كل الأرقام مختبرة تاريخيًا. لا ضمان ربح. والقرار لك.

الكود والنتائج:

- `r9_run.py` و`r10_bigmove.py` و`r10_options.py` و`r10_model.py`.
- `results/r9/` و`results/r10/`.

ملحق: النتائج الخام (results/r9 و results/r10)

r9/RESULTS.md

Return per trade: STOCK on notional; CALL/LONG on premium; SPREAD on net debit. Each leg pays a half-spread (ETF 1.5 %, stock 3 %) + $0.65/contract/side.

SQZ

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	all	+0.9% (n 36, t 1.47, win 53%)	+19.4% (n 34, t 1.10, win 47%)	-4.5% (n 34, t -0.40, win 47%)	+0.1% (n 20, t 0.02, win 55%)
orig	2024-02→2025-06	+0.7% (n 22, t 0.98, win 59%)	+15.3% (n 20, t 0.72, win 50%)	-5.1% (n 20, t -0.34, win 50%)	-1.4% (n 12, t -0.16, win 50%)
orig	2025-07→2026-10	+1.2% (n 14, t 1.08, win 43%)	+25.3% (n 14, t 0.81, win 43%)	-3.7% (n 14, t -0.20, win 43%)	+2.4% (n 8, t 0.32, win 62%)
orig	2× half-spread		+12.4% (n 34, t 0.75, win 47%)	-22.1% (n 34, t -2.40, win 35%)	-5.7% (n 20, t -1.01, win 55%)
new	all	+0.3% (n 79, t 0.88, win 53%)	+11.9% (n 73, t 1.06, win 48%)	-5.8% (n 72, t -0.72, win 50%)	-7.0% (n 37, t -1.53, win 46%)
new	2024-02→2025-06	+0.7% (n 47, t 1.24, win 57%)	+17.6% (n 44, t 1.17, win 52%)	-0.4% (n 44, t -0.04, win 52%)	-8.1% (n 20, t -1.22, win 45%)
new	2025-07→2026-10	-0.1% (n 32, t -0.22, win 47%)	+3.4% (n 29, t 0.20, win 41%)	-14.2% (n 28, t -1.20, win 46%)	-5.7% (n 17, t -0.89, win 47%)
new	2× half-spread		+5.4% (n 73, t 0.51, win 45%)	-20.8% (n 72, t -3.02, win 40%)	-12.4% (n 37, t -2.89, win 38%)

TDL

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	all	+0.4% (n 317, t 1.95, win 57%)	-4.2% (n 298, t -0.89, win 45%)	-9.9% (n 289, t -2.62, win 47%)	-2.6% (n 89, t -1.10, win 49%)
orig	2024-02→2025-06	+0.1% (n 153, t 0.52, win 55%)	-9.5% (n 144, t -1.36, win 41%)	-15.1% (n 139, t -2.67, win 42%)	-0.1% (n 42, t -0.02, win 57%)
orig	2025-07→2026-10	+0.5% (n 164, t 2.39, win 60%)	+0.8% (n 154, t 0.13, win 49%)	-5.2% (n 150, t -1.02, win 52%)	-4.9% (n 47, t -1.86, win 43%)
orig	2× half-spread		-9.2% (n 298, t -2.09, win 44%)	-20.5% (n 289, t -6.09, win 42%)	-7.9% (n 89, t -3.50, win 36%)
new	all	+0.3% (n 564, t 1.87, win 58%)	-10.5% (n 444, t -2.51, win 38%)	-15.0% (n 404, t -4.34, win 39%)	-10.5% (n 89, t -3.71, win 34%)
new	2024-02→2025-06	+0.1% (n 303, t 0.64, win 56%)	-9.6% (n 244, t -1.78, win 41%)	-15.4% (n 224, t -3.47, win 41%)	-13.5% (n 48, t -3.16, win 29%)
new	2025-07→2026-10	+0.4% (n 261, t 2.00, win 60%)	-11.6% (n 200, t -1.77, win 34%)	-14.4% (n 180, t -2.66, win 38%)	-6.9% (n 41, t -1.97, win 39%)
new	2× half-spread		-15.7% (n 444, t -3.99, win 36%)	-25.9% (n 404, t -8.49, win 31%)	-15.7% (n 89, t -5.91, win 26%)

FOMC

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	all	+0.2% (n 63, t 1.93, win 57%)	+4.2% (n 63, t 0.77, win 49%)	-3.2% (n 63, t -0.78, win 43%)	-0.5% (n 16, t -0.25, win 62%)

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	2024-02→2025-06	+0.4% (n 33, t 2.37, win 67%)	+13.9% (n 33, t 1.67, win 55%)	+5.1% (n 33, t 0.83, win 52%)	+2.3% (n 8, t 0.69, win 75%)
orig	2025-07→2026-10	-0.0% (n 30, t -0.06, win 47%)	-6.5% (n 30, t -1.05, win 43%)	-12.3% (n 30, t -2.48, win 33%)	-3.3% (n 8, t -1.68, win 50%)
orig	2× half-spread		+1.1% (n 63, t 0.21, win 48%)	-9.3% (n 63, t -2.45, win 25%)	-3.5% (n 16, t -1.78, win 19%)

S1

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	all	+2.6% (n 237, t 3.97, win 60%)	—	—	-1.3% (n 92, t -0.16, win 42%)
orig	2024-02→2025-06	+3.0% (n 114, t 3.92, win 67%)	—	—	+0.8% (n 37, t 0.08, win 49%)
orig	2025-07→2026-10	+2.2% (n 123, t 2.14, win 54%)	—	—	-2.7% (n 55, t -0.24, win 38%)
orig	2× half-spread		—	—	-6.6% (n 92, t -0.89, win 39%)
new	all	+3.4% (n 387, t 5.01, win 58%)	—	—	+0.3% (n 79, t 0.03, win 39%)
new	2024-02→2025-06	+3.8% (n 200, t 5.41, win 65%)	—	—	+0.4% (n 34, t 0.03, win 38%)
new	2025-07→2026-10	+3.0% (n 187, t 2.50, win 51%)	—	—	+0.2% (n 45, t 0.01, win 40%)
new	2× half-spread		—	—	-5.6% (n 79, t -0.57, win 34%)

S3

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	all	+0.8% (n 22, t 0.68, win 45%)	—	—	+4.6% (n 10, t 0.13, win 30%)
orig	2024-02→2025-06	+0.7% (n 10, t 0.38, win 50%)	—	—	-31.1% (n 5, t -3.37, win 20%)
orig	2025-07→2026-10	+0.9% (n 12, t 0.55, win 42%)	—	—	+40.3% (n 5, t 0.59, win 40%)
orig	2× half-spread		—	—	-1.0% (n 10, t -0.03, win 20%)
new	all	+0.4% (n 34, t 0.30, win 44%)	—	—	-3.8% (n 6, t -0.12, win 33%)
new	2024-02→2025-06	+2.8% (n 16, t 1.54, win 56%)	—	—	+45.8% (n 3, t 1.00, win 67%)
new	2025-07→2026-10	-1.8% (n 18, t -1.05, win 33%)	—	—	-53.5% (n 3, t -4.38, win 0%)
new	2× half-spread		—	—	-9.4% (n 6, t -0.33, win 33%)

Pre-registered acceptance check

signal	instrument	orig > 0	new > 0	pooled t > 1	both halves > 0	2× spreads > 0	verdict
SQZ	CALL	✅	✅	✅	✅	✅	**Paper candidate**
SQZ	SPREAD	❌	❌	❌	❌	❌	**reject**
SQZ	LONG	✅	❌	❌	❌	❌	**reject**
TDL	CALL	❌	❌	❌	❌	❌	**reject**
TDL	SPREAD	❌	❌	❌	❌	❌	**reject**
TDL	LONG	❌	❌	❌	❌	❌	**reject**
FOMC	CALL	✅	n/a	❌	❌	✅	**reject**
FOMC	SPREAD	❌	n/a	❌	❌	❌	**reject**
FOMC	LONG	❌	n/a	❌	❌	❌	**reject**
S1	CALL	❌	❌	❌	❌	❌	**reject**
S1	SPREAD	❌	❌	❌	❌	❌	**reject**
S1	LONG	❌	✅	❌	❌	❌	**reject**
S3	CALL	❌	❌	❌	❌	❌	**reject**
S3	SPREAD	❌	❌	❌	❌	❌	**reject**

signal	instrument	orig > 0	new > 0	pooled t > 1	both halves > 0	2× spreads > 0	verdict
S3	LONG	✅	❌	❌	❌	❌	**reject**

r10/BIGMOVE.md

UP ≥ 5% in 2 days

Base rate: discovery 5.98 %, validation 4.57 %, new symbols 4.61 %. Share of these moves that came with an earnings-like gap (≥1.5 ATR on ≥2.5× volume): 8 %.

Single indicators — lift = P(event | indicator) / P(event) (sorted by discovery lift; indicators true on ≥ 1 % of days)

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
20d return < -10%	5499	**2.94**	3.05	3.05	+0.36% / +0.48% / +0.41%
RSI14 < 30	1428	**2.60**	2.74	2.41	+0.81% / +0.57% / +0.60%
5d return < -5%	5906	**2.53**	2.74	2.70	+0.38% / +0.50% / +0.50%
SPY 5d return < -3%	4706	**2.13**	2.31	2.52	+0.45% / +0.45% / +0.52%
VIX > 25	12088	**2.06**	2.24	2.35	+0.30% / +0.36% / +0.31%
volume > 2x avg	1850	**1.86**	1.98	1.72	+0.23% / +0.33% / +0.13%
5d return > +5%	6969	**1.70**	1.93	1.60	+0.07% / +0.25% / -0.01%
VIX up > 15% today	1451	**1.69**	1.11	1.39	+0.48% / +0.02% / +0.17%
ATR% top 20% of year	10901	**1.68**	1.74	1.76	+0.17% / +0.18% / +0.17%
close < lower BB	2710	**1.65**	1.60	1.59	+0.54% / +0.46% / +0.28%
close < SMA200	18628	**1.64**	1.67	1.59	+0.16% / +0.26% / +0.17%
close < SMA20 - 1.5 ATR	7892	**1.61**	1.62	1.58	+0.39% / +0.32% / +0.30%
20d return > +10%	7690	**1.61**	1.81	1.46	+0.21% / +0.46% / +0.12%
volume > 1.5x avg	5435	**1.58**	1.67	1.57	+0.24% / +0.28% / +0.14%
Stoch %K < 20	8728	**1.56**	1.66	1.50	+0.34% / +0.36% / +0.32%
within 10% of 52w low	4978	**1.41**	1.63	1.48	+0.20% / +0.22% / +0.22%
RSI2 < 10	6291	**1.40**	1.46	1.36	+0.41% / +0.42% / +0.31%
BB width pct > 80%	11055	**1.29**	1.35	1.32	+0.21% / +0.24% / +0.19%
close in bottom 20% of range	11631	**1.29**	1.31	1.30	+0.37% / +0.41% / +0.31%
Elliott: up count, wave 2	3667	**1.27**	1.19	1.22	+0.07% / +0.35% / +0.17%
ADX > 25	21535	**1.19**	1.18	1.12	+0.20% / +0.26% / +0.16%
Elliott: up count, wave 3	2776	**1.11**	1.35	1.11	+0.12% / +0.29% / +0.14%
volume < 0.7x avg	10759	**1.09**	0.84	0.92	+0.11% / +0.27% / +0.11%
MACD hist < 0 & rising	12391	**1.03**	1.02	1.06	+0.09% / +0.21% / +0.16%
Monday	10275	**1.02**	1.02	1.09	+0.19% / +0.31% / +0.21%
MACD bull cross (3d)	6536	**1.01**	0.87	0.99	+0.16% / +0.26% / +0.11%
inside day	6498	**1.00**	1.04	0.99	+0.17% / +0.21% / +0.18%
60d return > SPY 60d	27700	**0.99**	0.97	0.94	+0.17% / +0.24% / +0.14%
MACD hist rising	26964	**0.99**	0.98	0.97	+0.11% / +0.21% / +0.11%
NR7	8440	**0.96**	0.88	0.93	+0.11% / +0.20% / +0.15%
Elliott: down count, wave 5	2134	**0.96**	1.28	0.95	+0.19% / +0.22% / +0.05%
Elliott: down count, wave 2/4/B	8537	**0.94**	0.95	1.03	+0.19% / +0.18% / +0.18%
Friday	10922	**0.94**	1.06	0.94	+0.17% / +0.32% / +0.14%
MACD hist > 0	27652	**0.94**	0.97	0.92	+0.12% / +0.23% / +0.10%

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
BB squeeze (width pct < 20%)	12764	**0.90**	0.81	0.85	+0.18% / +0.26% / +0.13%
close crossed above SMA50 (3d)	5614	**0.89**	0.87	0.95	+0.18% / +0.08% / +0.12%
Elliott: down count, wave 3	3745	**0.88**	0.90	0.78	+0.10% / +0.16% / +0.03%
close > SMA20	31034	**0.80**	0.83	0.79	+0.13% / +0.22% / +0.09%
RSI14 50-70	28024	**0.78**	0.77	0.77	+0.14% / +0.20% / +0.11%
close > upper BB	3571	**0.78**	1.00	0.75	-0.01% / +0.29% / +0.01%
OBV at 20-day high	7504	**0.77**	0.79	0.73	+0.07% / +0.18% / +0.07%
SMA50 rising	33391	**0.77**	0.78	0.78	+0.16% / +0.21% / +0.12%
Elliott: up count, wave 4	2518	**0.76**	0.81	0.83	+0.02% / +0.20% / +0.11%
ADX < 15	8235	**0.76**	0.84	0.92	+0.09% / +0.24% / +0.10%
close > SMA50	32549	**0.75**	0.76	0.75	+0.14% / +0.21% / +0.10%
MACD line > 0	32674	**0.75**	0.78	0.75	+0.15% / +0.22% / +0.11%
close in top 20% of range	13307	**0.75**	0.89	0.90	-0.04% / +0.13% / -0.00%
SMA200 rising	37371	**0.70**	0.84	0.72	+0.15% / +0.21% / +0.13%
SMA50 > SMA200	36029	**0.70**	0.85	0.75	+0.14% / +0.23% / +0.13%
EMA9 > EMA20 > EMA50	27538	**0.70**	0.74	0.69	+0.13% / +0.21% / +0.10%
close > 1.5 ATR above SMA20	12751	**0.69**	0.82	0.68	+0.08% / +0.22% / +0.04%
RSI2 > 90	8402	**0.69**	0.80	0.73	+0.00% / +0.17% / +0.01%
SPY > SMA200	38415	**0.69**	0.78	0.70	+0.15% / +0.22% / +0.13%
close > SMA200	35935	**0.67**	0.78	0.69	+0.14% / +0.22% / +0.12%
RSI14 > 70	3977	**0.66**	0.92	0.62	+0.09% / +0.23% / -0.01%
ATR% bottom 20% of year	15986	**0.66**	0.61	0.58	+0.18% / +0.23% / +0.10%
Elliott: up count, wave 5	1950	**0.65**	0.63	0.95	-0.09% / +0.02% / +0.12%
Stoch %K > 80	15014	**0.65**	0.68	0.66	+0.07% / +0.17% / +0.04%
Elliott: up count, wave C	1511	**0.59**	0.54	0.69	+0.09% / +0.18% / +0.09%
VIX < 15	15796	**0.45**	0.57	0.38	+0.11% / +0.24% / +0.10%
within 3% of 52w high	11704	**0.27**	0.33	0.34	+0.05% / +0.15% / +0.06%

Best combinations found on DISCOVERY (stable in both halves 1999–2007 / 2008–2015), then checked out of sample — 938 stable combos found

combination	days DISC	lift DISC	lift VALID	lift NEW	P(event) VALID / NEW	mean 2d return DISC / VALID / NEW
BB width pct > 80% + 20d return < -10% + SPY 5d return < -3%	781	**4.84**	4.77	5.28	21.8% / 24.4%	+1.53% / +1.02% / +1.38%
ATR% top 20% of year + 20d return < -10% + VIX up > 15% today	208	**4.82**	3.08	3.92	14.1% / 18.1%	+1.53% / -0.44% / +0.27%
ATR% top 20% of year + 20d return < -10% + SPY 5d return < -3%	940	**4.77**	4.96	5.45	22.7% / 25.2%	+1.25% / +0.88% / +1.36%
RSI14 < 30 + volume > 1.5x avg + VIX > 25	222	**4.75**	3.68	4.30	16.8% / 19.8%	+1.89% / +0.32% / +0.80%
20d return < -10% + VIX > 25 + VIX up > 15% today	217	**4.70**	3.49	4.08	16.0% / 18.8%	+1.61% / -0.49% / +0.19%
RSI14 < 30 + 20d return < -10% + SPY 5d return < -3%	426	**4.55**	4.31	4.73	19.7% / 21.8%	+1.47% / +1.07% / +1.26%
RSI14 < 30 + ATR% top 20% of year + VIX > 25	498	**4.53**	4.76	4.95	21.8% / 22.8%	+1.87% / +0.97% / +1.45%
RSI14 < 30 + ATR% top 20% of year + SPY 5d return < -3%	384	**4.53**	4.30	4.86	19.7% / 22.4%	+1.93% / +1.05% / +1.54%
volume > 1.5x avg + 20d return < -10% + SPY 5d return < -3%	329	**4.52**	3.32	4.76	15.2% / 21.9%	+1.34% / +0.03% / +0.88%

combination	days DISC	lift DISC	lift VALID	lift NEW	P(event) VALID / NEW	mean 2d return DISC / VALID / NEW
RSI14 < 30 + 20d return < -10% + VIX > 25	573	**4.52**	4.73	4.62	21.6% / 21.3%	+1.55% / +1.01% / +1.19%

UP ≥ 2ATR in 2 days

Base rate: discovery 2.58 %, validation 3.07 %, new symbols 2.68 %. Share of these moves that came with an earnings-like gap (≥1.5 ATR on ≥2.5× volume): 14 %.

Single indicators — lift = P(event | indicator) / P(event) (sorted by discovery lift; indicators true on ≥ 1 % of days)

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
volume > 2x avg	1850	**2.27**	2.45	1.80	+0.23% / +0.33% / +0.13%
volume > 1.5x avg	5435	**1.89**	1.94	1.70	+0.24% / +0.28% / +0.14%
VIX up > 15% today	1451	**1.77**	1.30	1.49	+0.48% / +0.02% / +0.17%
close < lower BB	2710	**1.49**	1.39	1.24	+0.54% / +0.46% / +0.28%
RSI14 < 30	1428	**1.41**	1.74	1.32	+0.81% / +0.57% / +0.60%
Elliott: up count, wave C	1511	**1.36**	0.94	0.94	+0.09% / +0.18% / +0.09%
20d return > +10%	7690	**1.30**	1.33	1.06	+0.21% / +0.46% / +0.12%
ATR% bottom 20% of year	15986	**1.29**	1.25	1.20	+0.18% / +0.23% / +0.10%
RSI2 < 10	6291	**1.27**	1.33	1.14	+0.41% / +0.42% / +0.31%
close in bottom 20% of range	11631	**1.26**	1.17	1.11	+0.37% / +0.41% / +0.31%
RSI14 > 70	3977	**1.21**	1.25	1.04	+0.09% / +0.23% / -0.01%
SPY 5d return < -3%	4706	**1.19**	0.97	1.21	+0.45% / +0.45% / +0.52%
BB squeeze (width pct < 20%)	12764	**1.19**	1.28	1.22	+0.18% / +0.26% / +0.13%
60d return > SPY 60d	27700	**1.10**	1.05	1.02	+0.17% / +0.24% / +0.14%
within 10% of 52w low	4978	**1.08**	1.03	1.03	+0.20% / +0.22% / +0.22%
SPY > SMA200	38415	**1.08**	1.04	1.05	+0.15% / +0.22% / +0.13%
Stoch %K < 20	8728	**1.08**	1.16	1.04	+0.34% / +0.36% / +0.32%
close < SMA20 - 1.5 ATR	7892	**1.08**	1.02	1.02	+0.39% / +0.32% / +0.30%
EMA9 > EMA20 > EMA50	27538	**1.08**	1.06	1.01	+0.13% / +0.21% / +0.10%
Elliott: up count, wave 3	2776	**1.08**	1.10	1.06	+0.12% / +0.29% / +0.14%
MACD line > 0	32674	**1.06**	1.04	1.03	+0.15% / +0.22% / +0.11%
SMA50 rising	33391	**1.06**	1.06	1.05	+0.16% / +0.21% / +0.12%
ADX > 25	21535	**1.05**	1.05	0.99	+0.20% / +0.26% / +0.16%
close > upper BB	3571	**1.04**	1.25	1.05	-0.01% / +0.29% / +0.01%
Elliott: down count, wave 5	2134	**1.04**	0.63	0.76	+0.19% / +0.22% / +0.05%
Elliott: down count, wave 3	3745	**1.04**	1.14	0.87	+0.10% / +0.16% / +0.03%
close > SMA50	32549	**1.03**	1.01	1.03	+0.14% / +0.21% / +0.10%
OBV at 20-day high	7504	**1.03**	0.99	1.11	+0.07% / +0.18% / +0.07%
inside day	6498	**1.03**	1.02	0.97	+0.17% / +0.21% / +0.18%
close > SMA200	35935	**1.02**	1.01	1.01	+0.14% / +0.22% / +0.12%
SMA200 rising	37371	**1.02**	1.00	0.99	+0.15% / +0.21% / +0.13%
VIX < 15	15796	**1.02**	1.04	0.99	+0.11% / +0.24% / +0.10%
5d return > +5%	6969	**1.01**	1.30	0.97	+0.07% / +0.25% / -0.01%
SMA50 > SMA200	36029	**1.01**	1.00	0.99	+0.14% / +0.23% / +0.13%
5d return < -5%	5906	**1.01**	0.99	0.99	+0.38% / +0.50% / +0.50%
close > 1.5 ATR above SMA20	12751	**1.01**	1.00	0.98	+0.08% / +0.22% / +0.04%

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
Elliott: down count, wave 2/4/B	8537	**1.00**	0.83	0.96	+0.19% / +0.18% / +0.18%
ADX < 15	8235	**1.00**	1.07	1.10	+0.09% / +0.24% / +0.10%
close > SMA20	31034	**0.99**	0.96	1.00	+0.13% / +0.22% / +0.09%
RSI14 50-70	28024	**0.98**	0.93	1.01	+0.14% / +0.20% / +0.11%
MACD hist > 0	27652	**0.97**	0.98	0.96	+0.12% / +0.23% / +0.10%
close < SMA200	18628	**0.96**	0.96	0.98	+0.16% / +0.26% / +0.17%
MACD bull cross (3d)	6536	**0.94**	0.91	1.00	+0.16% / +0.26% / +0.11%
MACD hist rising	26964	**0.93**	0.94	1.00	+0.11% / +0.21% / +0.11%
close crossed above SMA50 (3d)	5614	**0.92**	0.85	1.04	+0.18% / +0.08% / +0.12%
Elliott: up count, wave 4	2518	**0.91**	0.98	0.97	+0.02% / +0.20% / +0.11%
within 3% of 52w high	11704	**0.91**	0.96	0.99	+0.05% / +0.15% / +0.06%
Stoch %K > 80	15014	**0.90**	0.90	0.94	+0.07% / +0.17% / +0.04%
Monday	10275	**0.89**	0.96	1.08	+0.19% / +0.31% / +0.21%
Friday	10922	**0.87**	0.94	0.91	+0.17% / +0.32% / +0.14%
NR7	8440	**0.87**	0.71	0.83	+0.11% / +0.20% / +0.15%
MACD hist < 0 & rising	12391	**0.87**	0.92	1.02	+0.09% / +0.21% / +0.16%
VIX > 25	12088	**0.86**	1.08	0.94	+0.30% / +0.36% / +0.31%
RSI2 > 90	8402	**0.85**	0.99	0.97	+0.00% / +0.17% / +0.01%
volume < 0.7x avg	10759	**0.83**	0.61	0.70	+0.11% / +0.27% / +0.11%
20d return < -10%	5499	**0.83**	0.74	0.92	+0.36% / +0.48% / +0.41%
Elliott: up count, wave 2	3667	**0.81**	1.08	0.98	+0.07% / +0.35% / +0.17%
close in top 20% of range	13307	**0.81**	0.83	0.90	-0.04% / +0.13% / -0.00%
BB width pct > 80%	11055	**0.79**	0.86	0.76	+0.21% / +0.24% / +0.19%
ATR% top 20% of year	10901	**0.74**	0.63	0.71	+0.17% / +0.18% / +0.17%
Elliott: up count, wave 5	1950	**0.56**	0.55	1.14	-0.09% / +0.02% / +0.12%

Best combinations found on DISCOVERY (stable in both halves 1999–2007 / 2008–2015), then checked out of sample — 64 stable combos found

combination	days DISC	lift DISC	lift VALID	lift NEW	P(event) VALID / NEW	mean 2d return DISC / VALID / NEW
volume > 1.5x avg + close in bottom 20% of range + SPY 5d return < -3%	350	**3.66**	1.67	2.06	5.1% / 5.5%	+1.01% / +0.44% / +0.60%
volume > 2x avg + volume > 1.5x avg + SPY 5d return < -3%	319	**3.53**	3.55	2.01	10.9% / 5.4%	+1.14% / +0.85% / +0.39%
volume > 2x avg + SPY 5d return < -3%	319	**3.53**	3.55	2.01	10.9% / 5.4%	+1.14% / +0.85% / +0.39%
ATR% bottom 20% of year + volume > 1.5x avg + 20d return > +10%	343	**3.28**	3.62	1.78	11.1% / 4.8%	+0.38% / +1.42% / -0.06%
volume > 1.5x avg + SPY 5d return < -3% + VIX up > 15% today	213	**3.28**	2.06	2.16	6.3% / 5.8%	+1.20% / +0.14% / +0.41%
close < lower BB + volume > 1.5x avg + SPY 5d return < -3%	441	**3.17**	2.64	2.19	8.1% / 5.9%	+1.37% / +0.62% / +0.69%
volume > 1.5x avg + close in bottom 20% of range + VIX up > 15% today	255	**2.89**	2.56	1.70	7.9% / 4.6%	+0.58% / +0.35% / -0.12%
close < lower BB + volume > 2x avg + close in bottom 20% of range	209	**2.79**	1.92	1.54	5.9% / 4.1%	+0.43% / +0.28% / +0.01%
volume > 2x avg + volume > 1.5x avg + close in bottom 20% of range	467	**2.74**	2.34	1.55	7.2% / 4.2%	+0.47% / +0.59% / +0.16%
volume > 2x avg + close in bottom 20% of range	467	**2.74**	2.34	1.55	7.2% / 4.2%	+0.47% / +0.59% / +0.16%

DOWN ≥ 5% in 2 days

Base rate: discovery 5.25 %, validation 3.67 %, new symbols 4.10 %. Share of these moves that came with an earnings-like gap (≥1.5 ATR on ≥2.5× volume): 9 %.

Single indicators — lift = P(event | indicator) / P(event) (sorted by discovery lift; indicators true on ≥ 1 % of days)

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
20d return < -10%	5499	**2.73**	2.65	2.68	-0.36% / -0.48% / -0.41%
5d return < -5%	5906	**2.37**	2.38	2.18	-0.38% / -0.50% / -0.50%
volume > 2x avg	1850	**2.02**	1.82	1.84	-0.23% / -0.33% / -0.13%
RSI14 < 30	1428	**2.00**	1.87	1.75	-0.81% / -0.57% / -0.60%
VIX > 25	12088	**1.99**	2.22	2.17	-0.30% / -0.36% / -0.31%
SPY 5d return < -3%	4706	**1.94**	2.19	2.02	-0.45% / -0.45% / -0.52%
5d return > +5%	6969	**1.88**	1.83	1.88	-0.07% / -0.25% / +0.01%
ATR% top 20% of year	10901	**1.81**	2.01	1.84	-0.17% / -0.18% / -0.17%
20d return > +10%	7690	**1.61**	1.48	1.55	-0.21% / -0.46% / -0.12%
close < SMA200	18628	**1.58**	1.60	1.52	-0.16% / -0.26% / -0.17%
volume > 1.5x avg	5435	**1.58**	1.71	1.65	-0.24% / -0.28% / -0.14%
Elliott: up count, wave 2	3667	**1.50**	0.98	1.20	-0.07% / -0.35% / -0.17%
VIX up > 15% today	1451	**1.43**	1.48	1.45	-0.48% / -0.02% / -0.17%
close < SMA20 - 1.5 ATR	7892	**1.39**	1.51	1.39	-0.39% / -0.32% / -0.30%
close < lower BB	2710	**1.39**	1.23	1.42	-0.54% / -0.46% / -0.28%
Stoch %K < 20	8728	**1.34**	1.42	1.24	-0.34% / -0.36% / -0.32%
BB width pct > 80%	11055	**1.34**	1.36	1.27	-0.21% / -0.24% / -0.19%
within 10% of 52w low	4978	**1.29**	1.60	1.25	-0.20% / -0.22% / -0.22%
RSI2 < 10	6291	**1.16**	1.17	1.10	-0.41% / -0.42% / -0.31%
ADX > 25	21535	**1.16**	1.17	1.10	-0.20% / -0.26% / -0.16%
volume < 0.7x avg	10759	**1.10**	0.75	0.91	-0.11% / -0.27% / -0.11%
MACD hist < 0 & rising	12391	**1.07**	1.02	1.03	-0.09% / -0.21% / -0.16%
close in bottom 20% of range	11631	**1.06**	0.99	1.02	-0.37% / -0.41% / -0.31%
close > upper BB	3571	**1.05**	0.85	0.94	+0.01% / -0.29% / -0.01%
60d return > SPY 60d	27700	**1.02**	1.01	0.96	-0.17% / -0.24% / -0.14%
MACD hist rising	26964	**1.01**	0.98	1.01	-0.11% / -0.21% / -0.11%
MACD bull cross (3d)	6536	**1.01**	0.79	1.00	-0.16% / -0.26% / -0.11%
close in top 20% of range	13307	**1.00**	1.00	1.10	+0.04% / -0.13% / +0.00%
Elliott: down count, wave 5	2134	**0.98**	1.42	1.15	-0.19% / -0.22% / -0.05%
Monday	10275	**0.98**	0.76	0.90	-0.19% / -0.31% / -0.21%
Elliott: up count, wave 3	2776	**0.98**	1.23	1.17	-0.12% / -0.29% / -0.14%
Elliott: down count, wave 2/4/B	8537	**0.97**	1.09	0.96	-0.19% / -0.18% / -0.18%
NR7	8440	**0.96**	0.91	0.92	-0.11% / -0.20% / -0.15%
inside day	6498	**0.95**	1.13	0.93	-0.17% / -0.21% / -0.18%
MACD hist > 0	27652	**0.95**	0.93	0.97	-0.12% / -0.23% / -0.10%
close crossed above SMA50 (3d)	5614	**0.93**	1.17	1.07	-0.18% / -0.08% / -0.12%
Elliott: down count, wave 3	3745	**0.92**	0.98	0.89	-0.10% / -0.16% / -0.03%
OBV at 20-day high	7504	**0.89**	0.84	0.83	-0.07% / -0.18% / -0.07%
Friday	10922	**0.88**	0.82	0.91	-0.17% / -0.32% / -0.14%

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
RSI2 > 90	8402	**0.86**	0.78	0.86	-0.00% / -0.17% / -0.01%
Elliott: up count, wave 4	2518	**0.86**	0.71	0.85	-0.02% / -0.20% / -0.11%
Elliott: up count, wave 5	1950	**0.83**	0.94	0.97	+0.09% / -0.02% / -0.12%
BB squeeze (width pct < 20%)	12764	**0.83**	0.69	0.80	-0.18% / -0.26% / -0.13%
ADX < 15	8235	**0.81**	0.73	0.92	-0.09% / -0.24% / -0.10%
close > SMA20	31034	**0.80**	0.81	0.84	-0.13% / -0.22% / -0.09%
SMA50 rising	33391	**0.78**	0.80	0.79	-0.16% / -0.21% / -0.12%
RSI14 50-70	28024	**0.77**	0.77	0.83	-0.14% / -0.20% / -0.11%
close > SMA50	32549	**0.77**	0.78	0.80	-0.14% / -0.21% / -0.10%
MACD line > 0	32674	**0.76**	0.76	0.80	-0.15% / -0.22% / -0.11%
close > 1.5 ATR above SMA20	12751	**0.75**	0.75	0.80	-0.08% / -0.22% / -0.04%
RSI14 > 70	3977	**0.75**	0.84	0.70	-0.09% / -0.23% / +0.01%
SMA50 > SMA200	36029	**0.74**	0.86	0.76	-0.14% / -0.23% / -0.13%
SMA200 rising	37371	**0.73**	0.88	0.75	-0.15% / -0.21% / -0.13%
EMA9 > EMA20 > EMA50	27538	**0.71**	0.73	0.73	-0.13% / -0.21% / -0.10%
Stoch %K > 80	15014	**0.71**	0.74	0.79	-0.07% / -0.17% / -0.04%
close > SMA200	35935	**0.70**	0.80	0.72	-0.14% / -0.22% / -0.12%
Elliott: up count, wave C	1511	**0.67**	0.63	0.84	-0.09% / -0.18% / -0.09%
SPY > SMA200	38415	**0.64**	0.75	0.68	-0.15% / -0.22% / -0.13%
ATR% bottom 20% of year	15986	**0.52**	0.47	0.53	-0.18% / -0.23% / -0.10%
VIX < 15	15796	**0.36**	0.51	0.34	-0.11% / -0.24% / -0.10%
within 3% of 52w high	11704	**0.29**	0.37	0.37	-0.05% / -0.15% / -0.06%

Best combinations found on DISCOVERY (stable in both halves 1999–2007 / 2008–2015), then checked out of sample — 774 stable combos found

combination	days DISC	lift DISC	lift VALID	lift NEW	P(event) VALID / NEW	mean 2d return DISC / VALID / NEW
BB width pct > 80% + 5d return > +5% + Elliott: up count, wave 2	235	**4.14**	1.55	2.20	5.7% / 9.0%	+1.22% / -0.29% / +0.14%
ATR% top 20% of year + 20d return < -10% + Elliott: up count, wave 2	239	**3.99**	2.24	4.04	8.2% / 16.6%	+0.19% / -0.81% / -0.29%
5d return > +5% + 20d return < -10% + VIX > 25	298	**3.90**	5.44	4.86	20.0% / 19.9%	-0.48% / +0.67% / +0.02%
20d return < -10% + VIX > 25 + Elliott: up count, wave 2	240	**3.89**	3.49	4.32	12.8% / 17.7%	-0.18% / -1.30% / -0.24%
ATR% top 20% of year + 5d return < -5% + Elliott: up count, wave 2	211	**3.79**	3.89	3.12	14.3% / 12.8%	-0.31% / -0.47% / -0.49%
5d return > +5% + 20d return > +10% + Elliott: up count, wave 2	279	**3.76**	1.50	2.24	5.5% / 9.2%	+0.83% / -0.34% / +0.03%
close < SMA200 + 20d return < -10% + Elliott: up count, wave 2	336	**3.69**	1.88	3.55	6.9% / 14.6%	+0.07% / -0.86% / -0.18%
20d return < -10% + Elliott: up count, wave 2	342	**3.62**	1.94	3.51	7.1% / 14.4%	+0.06% / -0.76% / -0.18%
BB width pct > 80% + 5d return > +5% + VIX > 25	623	**3.61**	4.49	3.66	16.5% / 15.0%	+0.49% / +0.72% / +0.59%
5d return > +5% + VIX > 25 + Elliott: up count, wave 2	295	**3.55**	4.19	3.81	15.4% / 15.6%	-0.02% / +0.26% / +0.62%

DOWN ≥ 2ATR in 2 days

Base rate: discovery 2.35 %, validation 2.88 %, new symbols 2.48 %. Share of these moves that came with an earnings-like gap (≥1.5 ATR on ≥2.5× volume): 17 %.

Single indicators — lift = P(event | indicator) / P(event) (sorted by discovery lift; indicators true on ≥ 1 % of days)

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
VIX up > 15% today	1451	**2.23**	1.18	1.73	-0.48% / -0.02% / -0.17%
volume > 2x avg	1850	**2.10**	1.81	1.88	-0.23% / -0.33% / -0.13%
volume > 1.5x avg	5435	**1.87**	1.76	1.85	-0.24% / -0.28% / -0.14%
Elliott: up count, wave C	1511	**1.49**	1.05	1.11	-0.09% / -0.18% / -0.09%
close < lower BB	2710	**1.41**	0.96	1.34	-0.54% / -0.46% / -0.28%
within 3% of 52w high	11704	**1.39**	1.12	1.10	-0.05% / -0.15% / -0.06%
SPY 5d return < -3%	4706	**1.22**	1.33	1.32	-0.45% / -0.45% / -0.52%
MACD hist < 0 & rising	12391	**1.21**	1.09	1.05	-0.09% / -0.21% / -0.16%
RSI14 < 30	1428	**1.19**	1.25	1.18	-0.81% / -0.57% / -0.60%
BB squeeze (width pct < 20%)	12764	**1.18**	1.03	1.15	-0.18% / -0.26% / -0.13%
RSI14 > 70	3977	**1.15**	1.27	1.07	-0.09% / -0.23% / +0.01%
60d return > SPY 60d	27700	**1.13**	1.05	1.04	-0.17% / -0.24% / -0.14%
SMA50 rising	33391	**1.12**	1.08	1.06	-0.16% / -0.21% / -0.12%
SMA50 > SMA200	36029	**1.12**	1.05	1.07	-0.14% / -0.23% / -0.13%
Elliott: down count, wave 3	3745	**1.11**	1.03	1.09	-0.10% / -0.16% / -0.03%
SMA200 rising	37371	**1.11**	1.04	1.07	-0.15% / -0.21% / -0.13%
close > upper BB	3571	**1.11**	0.89	1.08	+0.01% / -0.29% / -0.01%
EMA9 > EMA20 > EMA50	27538	**1.10**	1.07	1.05	-0.13% / -0.21% / -0.10%
close > SMA200	35935	**1.09**	1.03	1.06	-0.14% / -0.22% / -0.12%
Elliott: up count, wave 4	2518	**1.08**	0.85	0.96	-0.02% / -0.20% / -0.11%
close in top 20% of range	13307	**1.08**	1.10	1.07	+0.04% / -0.13% / +0.00%
VIX < 15	15796	**1.07**	0.95	0.96	-0.11% / -0.24% / -0.10%
close > SMA50	32549	**1.05**	1.03	1.03	-0.14% / -0.21% / -0.10%
MACD line > 0	32674	**1.05**	1.03	1.03	-0.15% / -0.22% / -0.11%
Elliott: up count, wave 5	1950	**1.05**	1.09	0.99	+0.09% / -0.02% / -0.12%
Elliott: down count, wave 2/4/B	8537	**1.05**	0.97	0.99	-0.19% / -0.18% / -0.18%
ATR% bottom 20% of year	15986	**1.05**	1.03	1.11	-0.18% / -0.23% / -0.10%
MACD hist rising	26964	**1.04**	0.96	0.96	-0.11% / -0.21% / -0.11%
ATR% top 20% of year	10901	**1.04**	1.04	0.96	-0.17% / -0.18% / -0.17%
SPY > SMA200	38415	**1.03**	0.96	1.00	-0.15% / -0.22% / -0.13%
RSI14 50-70	28024	**1.02**	0.93	0.98	-0.14% / -0.20% / -0.11%
OBV at 20-day high	7504	**1.01**	1.04	1.00	-0.07% / -0.18% / -0.07%
MACD bull cross (3d)	6536	**1.00**	0.62	0.94	-0.16% / -0.26% / -0.11%
Stoch %K > 80	15014	**1.00**	0.96	0.95	-0.07% / -0.17% / -0.04%
inside day	6498	**0.99**	1.01	0.92	-0.17% / -0.21% / -0.18%
close > SMA20	31034	**0.98**	0.95	0.97	-0.13% / -0.22% / -0.09%
close crossed above SMA50 (3d)	5614	**0.97**	1.03	1.03	-0.18% / -0.08% / -0.12%
close < SMA20 - 1.5 ATR	7892	**0.96**	1.11	1.09	-0.39% / -0.32% / -0.30%
ADX > 25	21535	**0.96**	1.13	1.00	-0.20% / -0.26% / -0.16%
ADX < 15	8235	**0.94**	0.97	1.01	-0.09% / -0.24% / -0.10%
RSI2 > 90	8402	**0.94**	0.86	0.94	-0.00% / -0.17% / -0.01%
5d return > +5%	6969	**0.94**	0.76	0.85	-0.07% / -0.25% / +0.01%
within 10% of 52w low	4978	**0.93**	0.97	1.01	-0.20% / -0.22% / -0.22%

indicator	days (disc.)	lift DISC	lift VALID	lift NEW	mean 2d return DISC / VALID / NEW
close > 1.5 ATR above SMA20	12751	**0.92**	0.95	0.92	-0.08% / -0.22% / -0.04%
RSI2 < 10	6291	**0.92**	1.10	1.06	-0.41% / -0.42% / -0.31%
20d return > +10%	7690	**0.91**	0.86	0.83	-0.21% / -0.46% / -0.12%
close in bottom 20% of range	11631	**0.89**	0.98	0.99	-0.37% / -0.41% / -0.31%
MACD hist > 0	27652	**0.88**	0.87	0.91	-0.12% / -0.23% / -0.10%
20d return < -10%	5499	**0.88**	0.77	0.87	-0.36% / -0.48% / -0.41%
Stoch %K < 20	8728	**0.87**	1.01	1.00	-0.34% / -0.36% / -0.32%
BB width pct > 80%	11055	**0.86**	1.06	0.81	-0.21% / -0.24% / -0.19%
VIX > 25	12088	**0.85**	1.09	0.95	-0.30% / -0.36% / -0.31%
Elliott: up count, wave 3	2776	**0.84**	0.92	0.91	-0.12% / -0.29% / -0.14%
Monday	10275	**0.84**	0.75	0.84	-0.19% / -0.31% / -0.21%
Elliott: down count, wave 5	2134	**0.84**	0.78	0.99	-0.19% / -0.22% / -0.05%
close < SMA200	18628	**0.82**	0.92	0.89	-0.16% / -0.26% / -0.17%
5d return < -5%	5906	**0.79**	0.99	0.91	-0.38% / -0.50% / -0.50%
NR7	8440	**0.79**	0.85	0.74	-0.11% / -0.20% / -0.15%
volume < 0.7x avg	10759	**0.72**	0.53	0.62	-0.11% / -0.27% / -0.11%
Friday	10922	**0.71**	0.66	0.81	-0.17% / -0.32% / -0.14%
Elliott: up count, wave 2	3667	**0.65**	0.69	0.79	-0.07% / -0.35% / -0.17%

Best combinations found on DISCOVERY (stable in both halves 1999–2007 / 2008–2015), then checked out of sample — 218 stable combos found

combination	days DISC	lift DISC	lift VALID	lift NEW	P(event) VALID / NEW	mean 2d return DISC / VALID / NEW
SMA200 rising + close < lower BB + VIX up > 15% today	273	**4.52**	0.70	2.17	2.0% / 5.4%	-0.28% / -0.74% / -0.50%
SMA200 rising + volume > 1.5x avg + VIX up > 15% today	306	**3.90**	1.84	3.31	5.3% / 8.2%	-0.33% / -0.18% / -0.31%
SMA50 > SMA200 + volume > 1.5x avg + VIX up > 15% today	296	**3.89**	1.98	3.24	5.7% / 8.0%	-0.30% / -0.18% / -0.31%
BB squeeze (width pct < 20%) + volume > 2x avg + 60d return > SPY 60d	201	**3.81**	1.83	3.15	5.3% / 7.8%	+0.52% / -0.76% / +0.03%
MACD hist < 0 & rising + SMA50 rising + volume > 1.5x avg	461	**3.70**	2.69	2.13	7.7% / 5.3%	+0.09% / -0.08% / -0.36%
MACD hist < 0 & rising + volume > 1.5x avg + 60d return > SPY 60d	409	**3.64**	2.87	2.02	8.3% / 5.0%	+0.24% / +0.13% / -0.29%
SMA200 rising + 60d return > SPY 60d + VIX up > 15% today	571	**3.58**	1.40	1.99	4.0% / 4.9%	-0.06% / -0.13% / -0.22%
SMA50 rising + BB squeeze (width pct < 20%) + volume > 2x avg	251	**3.56**	2.26	2.87	6.5% / 7.1%	+0.32% / -0.43% / +0.15%
SMA50 > SMA200 + BB squeeze (width pct < 20%) + volume > 2x avg	240	**3.55**	1.95	2.47	5.6% / 6.1%	+0.41% / -0.77% / -0.06%
SMA50 rising + volume > 1.5x avg + Elliott: down count, wave 3	243	**3.51**	1.97	2.00	5.7% / 5.0%	-0.43% / -0.52% / +0.13%

r10/TOPCOMBO_SYMMETRY.md

set	days	P(up ≥5%)	P(down ≥5%)	base up / down	mean 2d	median 2d	P(2d > 0)
DISC	781	28.9%	15.0%	6.0% / 5.2%	+1.53%	+1.52%	59%
VALID	536	21.8%	11.0%	4.6% / 3.7%	+1.02%	+1.13%	60%
NEW	2418	24.4%	13.0%	4.6% / 4.1%	+1.38%	+1.27%	60%

Trades: 2935; distinct selloff episodes: 129.

group	trades	stock mean	stock win	call (model) mean	call win
all	2935	+1.14%	59%	+13.6%	49%
original 15	1025	+1.12%	58%	+12.5%	49%
new 30	1910	+1.15%	60%	+14.1%	49%
year 1999	21	+1.02%	52%	+0.5%	38%
year 2000	92	+1.40%	57%	+18.1%	51%
year 2001	190	+0.19%	52%	-1.7%	45%
year 2002	243	+0.82%	61%	+7.1%	54%
year 2003	8	+1.10%	50%	-0.9%	50%
year 2004	10	+1.72%	70%	+11.5%	30%
year 2005	5	+2.09%	80%	+44.6%	80%
year 2006	16	+2.41%	75%	+50.0%	56%
year 2007	53	+1.69%	58%	+29.7%	49%
year 2008	514	+1.71%	55%	+21.8%	47%
year 2009	100	+2.65%	63%	+28.3%	49%
year 2010	134	+0.73%	59%	+0.8%	39%
year 2011	184	+1.32%	67%	+23.5%	57%
year 2012	27	+1.12%	70%	+25.7%	56%
year 2013	3	+1.14%	67%	-9.1%	33%
year 2014	21	+1.09%	62%	+9.8%	48%
year 2015	53	+2.77%	89%	+43.9%	68%
year 2016	49	+1.71%	69%	+16.0%	55%
year 2018	158	+0.77%	63%	+15.6%	47%
year 2019	21	+2.25%	81%	+27.6%	71%
year 2020	502	+0.30%	54%	+6.6%	46%
year 2021	16	+0.89%	62%	+20.9%	62%
year 2022	277	+0.97%	57%	+3.9%	44%
year 2023	9	-0.99%	33%	-39.8%	0%
year 2024	43	+1.25%	67%	+10.6%	51%
year 2025	183	+1.92%	62%	+16.8%	52%
year 2026	3	+6.48%	100%	+206.2%	100%

By episode (equal weight per episode)

Episodes with positive mean stock return: 73% ; with positive mean call return: 56% ; mean of episode means: stock +1.61%, call +14.4%; t over episodes (stock) 6.81, (call) 3.42

episode start	trades	stock mean	call mean
1999-07-22	10	-1.19%	-37.4%
1999-08-31	2	+0.76%	+10.3%
1999-09-23	5	+1.45%	+8.7%
1999-10-13	4	+6.14%	+79.9%
2000-01-05	2	+5.91%	+58.5%
2000-01-26	3	+2.51%	+51.2%

episode start	trades	stock mean	call mean
2000-02-22	20	+0.23%	+0.7%
2000-04-13	11	+3.68%	+33.6%
2000-05-03	5	-1.03%	-33.1%
2000-05-22	7	-3.29%	-15.0%
2000-09-18	5	-1.90%	+87.2%
2000-10-10	12	+3.13%	+44.6%
2000-11-10	9	+4.32%	+16.8%
2000-12-18	18	+1.44%	+8.7%
2001-01-05	6	+4.61%	-7.3%
2001-02-08	1	+5.39%	+7.1%
2001-02-20	14	-0.55%	-14.5%
2001-03-12	45	+0.54%	-6.4%
2001-04-03	5	+4.81%	+76.8%
2001-05-29	1	+6.88%	+121.6%
2001-07-10	4	-0.46%	-9.5%
2001-07-24	2	+2.11%	-4.1%
2001-08-09	2	-1.52%	-43.5%
2001-08-31	109	-0.40%	-1.0%
2001-12-13	1	-1.33%	-48.0%
2002-02-04	2	+1.11%	-0.5%
2002-02-21	2	+5.01%	+35.8%
2002-04-26	5	+4.41%	+57.6%
2002-06-03	5	-0.64%	-17.4%
2002-06-24	21	-0.43%	-6.7%
2002-07-10	118	+0.38%	+4.6%
2002-08-05	32	+4.72%	+47.9%
2002-08-28	2	+3.03%	+35.9%
2002-09-17	55	-0.38%	-8.8%
2002-12-05	1	-2.85%	-55.6%
2003-01-30	5	+1.49%	+16.3%
2003-03-10	1	-1.45%	-70.1%
2003-06-25	1	+4.19%	+36.3%
2003-09-26	1	-1.40%	-55.2%
2004-03-11	4	-0.63%	-29.3%
2004-08-06	6	+3.29%	+38.8%
2005-04-15	4	+2.64%	+63.6%
2005-10-10	1	-0.08%	-31.6%
2006-05-17	12	+0.61%	+0.7%
2006-06-13	4	+7.83%	+197.7%
2007-03-05	1	+2.94%	+52.3%
2007-06-26	1	+1.69%	+3.9%
2007-07-26	7	-0.83%	-26.4%
2007-08-14	12	+0.05%	+10.9%
2007-10-19	1	+0.19%	-27.0%
2007-11-07	30	+2.98%	+53.8%
2007-12-17	1	+0.38%	-15.5%

episode start	trades	stock mean	call mean
2008-01-03	77	+0.96%	+11.6%
2008-02-08	9	+1.31%	+2.2%
2008-03-05	13	+0.23%	+36.3%
2008-05-23	1	+2.27%	-8.5%
2008-06-12	4	+2.38%	+22.9%
2008-06-23	18	-2.77%	-23.3%
2008-07-15	16	+11.13%	+222.2%
2008-09-04	31	+4.60%	+75.8%
2008-09-29	338	+1.27%	+12.0%
2008-12-04	5	+11.57%	+27.2%
2008-12-15	2	+8.55%	+19.3%
2009-01-16	8	+1.49%	+43.0%
2009-02-04	88	+2.78%	+28.0%
2009-05-15	1	+8.10%	+54.0%
2009-07-07	2	+0.78%	-6.4%
2009-10-28	1	-0.59%	-17.8%
2010-01-25	2	-0.53%	-29.1%
2010-05-06	107	+1.01%	+6.5%
2010-06-24	18	-0.62%	-23.7%
2010-08-13	6	+0.07%	-21.8%
2010-11-17	1	+1.74%	+25.9%
2011-03-15	4	+0.89%	+10.4%
2011-06-06	2	-0.61%	-37.0%
2011-07-28	137	+0.82%	+21.5%
2011-09-06	4	+1.11%	-12.6%
2011-09-22	21	+5.49%	+75.1%
2011-11-21	16	+0.56%	-6.9%
2012-05-09	11	+2.30%	+58.2%
2012-06-01	7	+1.87%	+22.1%
2012-10-24	3	+1.22%	+23.5%
2012-11-08	6	-1.99%	-28.5%
2013-04-18	1	+0.67%	-23.7%
2013-06-24	2	+1.37%	-1.8%
2014-01-27	1	+2.05%	-1.8%
2014-10-10	14	+1.32%	+18.1%
2014-12-12	6	+0.40%	-7.7%
2015-01-15	1	+1.26%	-12.9%
2015-06-29	1	+0.27%	-21.6%
2015-08-21	48	+2.90%	+47.4%
2015-09-28	2	+2.23%	+48.5%
2015-11-13	1	+2.00%	-15.1%
2016-01-06	32	+1.32%	+9.0%
2016-02-08	15	+2.19%	+19.3%
2016-06-27	2	+4.32%	+104.8%
2018-02-02	17	+1.08%	+14.0%
2018-03-22	13	-0.08%	-6.8%

episode start	trades	stock mean	call mean
2018-10-10	56	+1.13%	+30.2%
2018-11-14	18	+1.62%	+19.5%
2018-12-07	54	+0.23%	+5.1%
2019-05-13	1	+1.92%	+22.0%
2019-08-02	20	+2.26%	+27.8%
2020-02-24	491	+0.29%	+6.7%
2020-09-21	5	+1.95%	+14.9%
2020-10-28	6	-0.04%	-13.6%
2021-01-29	2	+3.26%	+76.8%
2021-09-30	7	-0.27%	-7.9%
2021-12-01	7	+1.37%	+33.8%
2022-01-19	88	-0.13%	-4.6%
2022-02-23	6	+7.74%	+94.5%
2022-03-07	15	+4.87%	+112.5%
2022-04-26	89	+1.41%	+2.3%
2022-06-09	14	+0.03%	-17.7%
2022-08-23	10	-0.71%	-36.6%
2022-09-16	36	+0.28%	-13.5%
2022-10-11	7	+3.61%	+54.0%
2022-11-04	3	+2.61%	-10.0%
2022-12-07	9	-0.20%	-25.5%
2023-03-10	3	-0.73%	-30.2%
2023-09-21	5	-0.52%	-39.5%
2023-10-26	1	-4.08%	-69.3%
2024-04-16	4	-0.47%	-26.6%
2024-08-05	33	+1.61%	+18.6%
2024-12-18	6	+0.42%	-8.6%
2025-02-26	86	+0.37%	-4.3%
2025-04-03	92	+3.65%	+40.3%
2025-11-18	5	-3.12%	-53.7%
2026-03-30	3	+6.48%	+206.2%

r10/OPTIONS.md

Rule: BB-width percentile > 80 % AND 20-day return < −10 % AND SPY 5-day return < −3 % → buy at 15:50, exit 15:50 two sessions later.

- **orig: stock only, all 5-minute history** — 421 trades, mean +0.75 %, median +1.05 %, win 59 %, t 2.68
- **new: stock only, all 5-minute history** — 165 trades, mean +2.46 %, median +1.87 %, win 71 %, t 5.98

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	2024-02→2026-10	+2.4% (n 73, t 3.30, win 66%)	+19.5% (n 64, t 1.71, win 48%)	+5.7% (n 63, t 0.70, win 52%)	+4.6% (n 4, t 1.47, win 75%)
orig	2024-02→2025-06	+2.4% (n 71, t 3.22, win 66%)	+20.8% (n 63, t 1.81, win 49%)	+6.6% (n 62, t 0.82, win 53%)	+4.6% (n 4, t 1.47, win 75%)

universe	period	STOCK	CALL (weekly)	SPREAD (weekly)	LONG (45–90 d, Δ0.75)
orig	2025-07→2026-10	—	—	—	—
orig	2× half-spread		+13.0% (n 64, t 1.21, win 48%)	-7.3% (n 63, t -1.01, win 44%)	-1.5% (n 4, t -0.50, win 25%)
orig	entries by month	2024-08: 11, 2025-02: 3, 2025-03: 28, 2025-04: 29, 2025-11: 1, 2026-03: 1			
new	2024-02→2026-10	+2.6% (n 155, t 6.06, win 73%)	+18.6% (n 106, t 1.83, win 45%)	+4.3% (n 81, t 0.62, win 46%)	-5.7% (n 11, t -0.63, win 27%)
new	2024-02→2025-06	+2.7% (n 149, t 6.24, win 74%)	+16.3% (n 100, t 1.87, win 46%)	+3.3% (n 78, t 0.50, win 46%)	-3.1% (n 10, t -0.32, win 30%)
new	2025-07→2026-10	-0.3% (n 6, t -0.11, win 33%)	+57.2% (n 6, t 0.50, win 33%)	+31.8% (n 3, t 0.33, win 33%)	—
new	2× half-spread		+11.7% (n 106, t 1.22, win 39%)	-9.5% (n 81, t -1.57, win 37%)	-11.2% (n 11, t -1.31, win 27%)
new	entries by month	2024-04: 3, 2024-08: 22, 2025-02: 1, 2025-03: 37, 2025-04: 86, 2025-11: 4, 2026-03: 2			

r10/REAL_ROBUST.md

cut	n	weekly CALL mean (real)	win
all (both universes)	170	+19.0%	46%
without 5 best	162	+9.3%	45%
without 10 best	154	+4.1%	44%
month 2024-04	3	+6.4%	33%
month 2024-08	28	+12.0%	54%
month 2025-02	3	-54.9%	0%
month 2025-03	52	+14.1%	50%
month 2025-04	77	+26.2%	45%
month 2025-11	5	-70.0%	0%
month 2026-03	2	+316.7%	100%
ETFs	10	-0.7%	30%
stocks	160	+20.2%	48%